HDOC DAY – 59

Algorithm

1. Start.
2. Read the string s.
3. Remove the newline character from the string.
4. Pass the address of s to longest_unique_substring().
5. Find the length of the string.
6. If the string is empty, display 0.
7. Initialize max_length to 0.
8. Start from each character of the string.
9. Check each following character with the characters already present in the current substring.
10. If a repeated character is found, stop the current substring.
11. Otherwise, increase current_length.
12. Update max_length if current_length is greater.
13. Display max_length.
14. Stop.

C PROGRAM:

```
#include <stdio.h>

#include <string.h>


void longest_unique_substring(char *s)

{

   int i, j, k;

   int current_length, max_length = 0;
```

```
int n = strlen(s);

if (n == 0)
{
   printf("0\n");
   return;
}

for (i = 0; i < n; i++)
{
   current_length = 0;

   for (j = i; j < n; j++)
   {
      for (k = i; k < j; k++)
      {
         if (*(s + k) == *(s + j))
            break;
      }

      if (k < j)
         break;

      current_length++;
```

```
            if (current_length > max_length)
                max_length = current_length;
        }
    }

    printf("Length of longest substring: %d\n", max_length);
}

int main()
{
    char s[100];

    printf("Enter the string: ");
    fgets(s, sizeof(s), stdin);

    s[strcspn(s, "\n")] = '\0';

    longest_unique_substring(s);

    return 0;
}
```

TEST CASE TABLE:

Test Case	Input String	Expected Output
1	abcabcbb	Length of longest substring: 3
2	bbbbb	Length of longest substring: 1

3	pwwkew	Length of longest substring: 3
4	Empty string	0
5	abcdef	Length of longest substring: 6
6	aAbBcC	Length of longest substring: 6